20/12/24 MLS

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[22CSE372]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

FUNDAMENTALS OF IOS APPLICATION DEVELOPMENT IN SWIFT

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A			**5x2=10**
1.	**a.**	List the different numeric types in Swift	L1	C1	**02**
	b.	Describe the function of Xcode in iOS app development.	L2	C2	**02**
	c.	List the controls available in UIKit for displaying data.	L1	C3	**02**
	d.	State the purpose of alignment rect in Auto Layout.	L1	C4	**02**
	e.	Explain the Model-View-Controller (MVC) design pattern.	L2	C5	**02**
		PART - B			**6x5=30**
2.	**a.**	Explain the structure of Swift's control transfer statements, including continue, break, and fall through.	L2	C1	**05**
		OR			
	b.	Describe the purpose and differences between if, switch, and for-in statements in Swift.	L2	C1	**05**
3.	**a.**	List the steps involved in setting up Xcode for iOS app.	L1	C1	**05**
		OR			
	b.	Explain the process of enrolling in the iOS developer program and state its key benefits.	L2	C1	**05**
4.	**a.**	Interpret the purpose of ToolBar and Bar Button Item in the context of building interactive user interfaces.	L3	C2	**05**
		OR			
	b.	Classify the key controls used for navigation in iOS apps.	L1	C2	**05**
5.	**a.**	Explain how to add subviews to an existing StackView and discuss the impact on app design.	L2	C3	**05**
		OR			
	b.	State the process of creating a StackView in code.	L2	C3	**05**
6.	**a.**	Describe the purpose of a Tab Bar Controller in an iOS.	L2	C4	**05**
		OR			
	b.	Classify the main components of the Model-View-Controller (MVC) design pattern.	L3	C4	**05**
7.	**a.**	Describe the different types of Mac OS, highlighting their key differences.	L2	C5	**05**
		OR			
	b.	Summarize the features of Swift as a programming language and compares to other programming languages.	L2	C5	**05**
		PART - C			**6x10=60**
8.	**a.**	Explain the architecture of iOS, focusing on its different layers Core OS, Core Services, Media, and Cocoa Touch.	L2	C1	**10**
		OR			

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	b.	Discuss the importance of control transfer statements (e.g., continue, break, fallthrough) in Swift.	L2	C1	**10**
9.	**a.**	Summarize the key features of Mac OS and classify the different types and versions of Mac OS.	L2	C1	**10**
		OR			
	b.	Classify the numeric data types in Swift, such as integer types and floating-point types.	L3	C1	**10**
10.	**a.**	Describe how classes and inheritance are used in Swift for organizing data.	L2	C2	**10**
		OR			
	b.	Explain the difference between structures and classes in Swift.	L2	C2	**10**
11.	**a.**	Summarize the usage of ImageView, Slider, and Page Control in UIKit.	L2	C3	**10**
		OR			
	b.	Classify the various views and controls provided by UIKit.	L2	C3	**10**
12.	**a.**	Implement an iOS app that uses Aspect Ratio Constraints to maintain the proportion of elements across different screen sizes.	L3	C4	**10**
		OR			
	b.	Illustrate how to hide and remove subviews from an existing StackView.	L3	C4	**10**
13.	**a.**	Explain the concept of Segues and Navigation Controllers in iOS.	L2	C5	**10**
		OR			
	b.	Describe the role of Tab Bar Controllers in iOS apps.	L2	C5	**10**